FORMULARIO

A continuación, encontrará algunas fórmulas que le pueden ser de utilidad durante la evaluación:

I. Sea el modelo de regresión lineal simple:

$$y_i = \alpha + \beta \cdot x_i + u_i$$

$\hat{\beta}_{MCO} = \frac{\sum(x_i - \bar{x}) \cdot (y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$ $\hat{\alpha} = \bar{y} - \hat{\beta}_{MCO} \cdot \bar{x}$	$Var(\hat{\beta}\|X) = \frac{\widehat{\sigma^2}}{\sum(x_i - \bar{x})^2}$ $Var(\hat{\alpha}\|X) = \frac{\widehat{\sigma^2} \sum x_i^2}{N \sum(x_i - \bar{x})^2}$ $\widehat{\sigma^2} = \frac{\sum \hat{u}_i{}^2}{N-2} = \frac{\sum(y_i - \hat{y}_i)^2}{N-2}$	$R^2 = \frac{SEC}{STC} = 1 - \frac{SRC}{STC}$ $STC = \sum(y_i - \bar{y})^2$ $SEC = \sum(\hat{y}_i - \bar{y})^2$ $SRC = \sum \hat{u}_i^2 = \sum(y_i - \hat{y}_i)^2$

$$\sum(x_i - \bar{x}) \cdot (y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$$

$$\sum(x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$$

II. Sea el modelo de regresión lineal múltiple:

$$Y = X\beta + U$$

$\hat{\beta}_{MCO} = (X'X)^{-1}X'Y$	$Var(\hat{\beta}_{MCO}\|X) = \sigma^2(X'X)^{-1}$ $\widehat{\sigma^2} = \frac{\hat{U}'\hat{U}}{N-(K+1)}$	$R^2 = \frac{SEC}{STC} = 1 - \frac{SRC}{STC}$ $STC = \sum(y_i - \bar{y})^2$ $SEC = \sum(\hat{y}_i - \bar{y})^2$ $SRC = \hat{U}'\hat{U} = \sum(y_i - \hat{y}_i)^2$

III. Tests de hipótesis:

En un modelo con $k+1$ parámetros estimados:

$$T_{obs} = \frac{\hat{\beta}_k - c}{sd(\hat{\beta}_k)} \sim t_{(N-k-1)}$$

En un modelo con q restricciones y $k+1$ parámetros estimados en el modelo original:

$$F = \frac{(SRC_r - SRC_{nr})/q}{SRC_{nr}/(n-k-1)} \sim F_{q,(n-k-1)} \qquad \text{ó} \qquad F = \frac{(R_{nr}^2 - R_r^2)/q}{(1-R_{nr}^2)/(n-(k+1))} \sim F_{q,(N-k-1)}$$